Akoizo Original Practice - Form 2 - Answer key

Division B | Food Science | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Why are saturated fats usually solid at room temperature?

Unit: Fats: saturation, esters, and emulsions.
Worked-case data: A team churns 150 mL of cream (mass 148 g) and collects 52 g of butter. If butter is about 80% fat, estimate the cream's fat percentage.
Separate investigation observations:
Trial 1: Cream at 4 °C: no butter after 15 minutes of shaking.
Trial 2: Cream at 13 °C: butter formed in 8 minutes.
Trial 3: Cream at 25 °C: soft, greasy butter; cloudy buttermilk.

A. They are water-soluble

B. They have fewer carbons

C. Straight chains pack tightly together

D. They contain double bonds

Correct: C

2. (2 points) Why does adding salt make an ice bath colder?

Unit: Freezing point depression and ice cream.
Worked-case data: Estimate the freezing point of meltwater in which 2.0 mol of NaCl dissolves per kilogram of water.
Separate investigation observations:
Trial 1: Milk, 30 mL salt: soft slush, 2 cm tall.
Trial 2: Milk, 90 mL salt: firmer, 4 cm tall.
Trial 3: Cream, 90 mL salt: firm and smooth, 7 cm tall.

A. Salt releases cold energy

B. Salt reacts with ice to form a cold gas

C. Salt raises the melting point

D. Salt lowers the freezing point, so more ice melts and absorbs heat

Correct: D

3. (2 points) Which vitamin helps the body absorb calcium?

Unit: Nutrition and food labels.
Worked-case data: A label lists 300 mg of calcium per serving and a Daily Value of 1,300 mg. A student drinks 3 servings. What percent DV is that, and is one serving “high”?
Separate investigation observations:
Milk A: 8 g fat, 12 g total sugars, 0 g added sugars, 25% DV calcium.
Milk B: 0 g fat, 12 g total sugars, 0 g added sugars, 25% DV calcium.
Milk C: 5 g fat, 24 g total sugars, 12 g added sugars, 25% DV calcium.

A. Vitamin D

B. Vitamin C

C. Vitamin K only

D. Riboflavin

Correct: A

4. (2 points) About what percentage of milk protein is casein?

Unit: What milk is made of: sugars, fats, and proteins.
Worked-case data: A 250 mL glass of whole milk weighs about 258 g. Using 3.3% protein and 4.8% lactose by mass, estimate the grams of protein and lactose.
Separate investigation observations:
Liquid A: Clear after acid is added and curds are removed.
Test of A: Contains lactose and dissolved proteins.
Solid B: White clumps that formed at pH 4.6.

A. 20%

B. 50%

C. 5%

D. 80%

Correct: D

5. (2 points) Which milk would likely give the greatest curd mass?

Unit: Proteins and curds.
Worked-case data: Milk in a beaker has mass 245.0 g; the empty beaker is 120.8 g. After curdling and draining, the curds on a 3.2 g cheesecloth weigh 22.0 g together. Find percent curd.
Separate investigation observations:
Trial A: Whole milk, 2 mL vinegar: 15% curd.
Trial B: Skim milk, 2 mL vinegar: 11% curd.
Trial C: Whole milk, 0.2 mL vinegar: 3% curd, cloudy whey.

A. Whole milk

B. Skim milk

C. Lactose-free skim milk

D. Water

Correct: A

6. (2 points) Which reaction gives browned, toasty flavors in heated dairy?

Unit: Processing, cooking, and preservation.
Worked-case data: A dairy heats milk to 72 °C for 15 seconds, homogenizes it, and stores it at 4 °C. Another heats to 140 °C for 3 seconds and packages it in sterile cartons. Compare the products.
Separate investigation observations:
Sample X: Raw milk left at room temperature for 2 days: sour smell, clumps.
Sample Y: HTST milk, refrigerated 2 days: fresh.
Sample Z: Unopened UHT carton at room temperature, 2 months: fresh.

A. Freezing point depression

B. The Maillard reaction

C. Esterification

D. Hydrolysis of lactose

Correct: B

7. (2 points) Why does lactose-free milk taste sweeter?

Unit: Lactose intolerance, allergies, alternatives, and the notebook.
Worked-case data: A student plans to test how vinegar volume affects curd mass. Design one notebook experiment.
Separate investigation observations:
Person 1: Bloating after milk; fine with yogurt and lactose-free milk.
Person 2: Hives within minutes of a tiny amount of cheese.
Test: Person 2 has antibodies to casein.

A. Fat is removed

B. Protein is concentrated

C. Lactose has been split into glucose and galactose

D. Sugar is added

Correct: C

8. (2 points) Butter is an example of which kind of emulsion?

Unit: Fats: saturation, esters, and emulsions.
Worked-case data: A team churns 150 mL of cream (mass 148 g) and collects 52 g of butter. If butter is about 80% fat, estimate the cream's fat percentage.
Separate investigation observations:
Trial 1: Cream at 4 °C: no butter after 15 minutes of shaking.
Trial 2: Cream at 13 °C: butter formed in 8 minutes.
Trial 3: Cream at 25 °C: soft, greasy butter; cloudy buttermilk.

A. Not an emulsion

B. Water-in-oil

C. Oil-in-water

D. Air-in-water

Correct: B

9. (2 points) Faster freezing of ice cream gives:

Unit: Freezing point depression and ice cream.
Worked-case data: Estimate the freezing point of meltwater in which 2.0 mol of NaCl dissolves per kilogram of water.
Separate investigation observations:
Trial 1: Milk, 30 mL salt: soft slush, 2 cm tall.
Trial 2: Milk, 90 mL salt: firmer, 4 cm tall.
Trial 3: Cream, 90 mL salt: firm and smooth, 7 cm tall.

A. No ice crystals

B. More lactose crystals only

C. Smaller ice crystals and a smoother texture

D. Larger ice crystals

Correct: C

10. (2 points) Fat-free milk has less than how much fat per serving?

Unit: Nutrition and food labels.
Worked-case data: A label lists 300 mg of calcium per serving and a Daily Value of 1,300 mg. A student drinks 3 servings. What percent DV is that, and is one serving “high”?
Separate investigation observations:
Milk A: 8 g fat, 12 g total sugars, 0 g added sugars, 25% DV calcium.
Milk B: 0 g fat, 12 g total sugars, 0 g added sugars, 25% DV calcium.
Milk C: 5 g fat, 24 g total sugars, 12 g added sugars, 25% DV calcium.

A. 5 g

B. 1 g

C. 3.25 g

D. 0.5 g

Correct: D

11. (4 points) Explain why milk is called an emulsion and a colloid.

Unit: What milk is made of: sugars, fats, and proteins.
Worked-case data: A 250 mL glass of whole milk weighs about 258 g. Using 3.3% protein and 4.8% lactose by mass, estimate the grams of protein and lactose.
Separate investigation observations:
Liquid A: Clear after acid is added and curds are removed.
Test of A: Contains lactose and dissolved proteins.
Solid B: White clumps that formed at pH 4.6.

answer: Fat exists as tiny droplets dispersed in water (an emulsion), and casein micelles are particles suspended in the liquid (a colloid). (4 points).

12. (4 points) Explain how acid causes casein micelles to clump.

Unit: Proteins and curds.
Worked-case data: Milk in a beaker has mass 245.0 g; the empty beaker is 120.8 g. After curdling and draining, the curds on a 3.2 g cheesecloth weigh 22.0 g together. Find percent curd.
Separate investigation observations:
Trial A: Whole milk, 2 mL vinegar: 15% curd.
Trial B: Skim milk, 2 mL vinegar: 11% curd.
Trial C: Whole milk, 0.2 mL vinegar: 3% curd, cloudy whey.

answer: Hydrogen ions neutralize negative charges and dissolve calcium phosphate. Near pH 4.6 micelles lose their repulsion and stick together. (4 points).

13. (4 points) Explain the difference between pasteurization and homogenization.

answer: Pasteurization uses heat to kill microbes for safety. Homogenization breaks fat globules so cream stays mixed; it does not affect safety. (4 points).

14. (4 points) Explain the difference between lactose intolerance and a milk allergy.

answer: Intolerance is a digestive problem caused by too little lactase; allergy is an immune reaction to milk proteins that can be life-threatening. (4 points).

15. (4 points) Why must cream be near 13 °C for efficient churning?

answer: At that temperature, fat globules are partly solid and partly liquid, so they stick and clump well; too cold or too warm reduces efficiency. (4 points).

16. (4 points) Give two ways to make ice cream stand taller in the lab.

Unit: Freezing point depression and ice cream.
Worked-case data: Estimate the freezing point of meltwater in which 2.0 mol of NaCl dissolves per kilogram of water.
Separate investigation observations:
Trial 1: Milk, 30 mL salt: soft slush, 2 cm tall.
Trial 2: Milk, 90 mL salt: firmer, 4 cm tall.
Trial 3: Cream, 90 mL salt: firm and smooth, 7 cm tall.

answer: Use cream with more fat for a firmer structure, and freeze faster in a colder salt-ice bath with vigorous shaking. (4 points).

17. (4 points) Describe one consequence of long-term calcium deficiency.

Unit: Nutrition and food labels.
Worked-case data: A label lists 300 mg of calcium per serving and a Daily Value of 1,300 mg. A student drinks 3 servings. What percent DV is that, and is one serving “high”?
Separate investigation observations:
Milk A: 8 g fat, 12 g total sugars, 0 g added sugars, 25% DV calcium.
Milk B: 0 g fat, 12 g total sugars, 0 g added sugars, 25% DV calcium.
Milk C: 5 g fat, 24 g total sugars, 12 g added sugars, 25% DV calcium.

answer: Weak, brittle bones (osteoporosis in adults), raising fracture risk. (4 points).

18. (4 points) Compare how casein and whey proteins respond to acid.

Unit: What milk is made of: sugars, fats, and proteins.
Worked-case data: A 250 mL glass of whole milk weighs about 258 g. Using 3.3% protein and 4.8% lactose by mass, estimate the grams of protein and lactose.
Separate investigation observations:
Liquid A: Clear after acid is added and curds are removed.
Test of A: Contains lactose and dissolved proteins.
Solid B: White clumps that formed at pH 4.6.

answer: Casein clumps at about pH 4.6 to form curds. Whey proteins stay dissolved and drain away in the whey. (4 points).